FIITJEE

JEE(Advanced)-2018

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS

FULL TEST – IX

PAPER-2

Q. No.	PHYSICS	Q. No.	CHEMISTRY	Q. No.	MATHEMATICS
1.	**C**	24.	**D**	47.	**B**
2.	**B**	25.	**A**	48.	**B**
3.	**C**	26.	**D**	49.	**C**
4.	**A, B, C, D**	27.	**A, B**	50.	**B, C, D**
5.	**B, C**	28.	**A, C, D**	51.	**C, D**
6.	**B, C**	29.	**A, B, C, D**	52.	**B, C**
7.	**A, B, C**	30.	**A, C, D**	53.	**A, C**
8.	**B, C, D**	31.	**A, B, C**	54.	**A, C**
9.	**B**	32.	**A**	55.	**B**
10.	**D**	33.	**C**	56.	**B**
11.	**5**	34.	**6**	57.	**1**
12.	**1**	35.	**4**	58.	**3**
13.	**8**	36.	**2**	59.	**2**
14.	**4**	37.	**3**	60.	**4**
15.	**5**	38.	**4**	61.	**3**
16.	**9**	39.	**5**	62.	**6**
17.	**4**	40.	**4**	63.	**3**
18.	**4**	41.	**3**	64.	**5**
19.	**4**	42.	**4**	65.	**6**
20.	**7**	43.	**4**	66.	**7**
21.	**00001.25**	44.	**00001.52**	67.	**00010.66**
22.	**00062.50**	45.	**00019.55**	68.	**00051.00**
23.	**00001.60**	46.	**00000.96**	69.	**00004.66**

Physics

PART – I

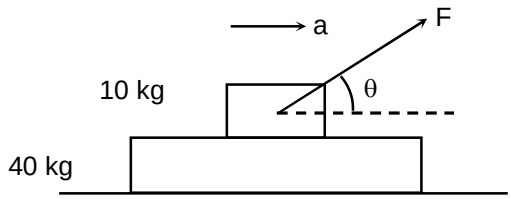
SECTION – A

1. An induced current will be developed in the loop due to change in flux.

2. Centre of mass will be at rest

3. Speed of pulley Y is 10 m/s and speed of block A is 10 m/s so speed of block B will be 10 m/s.

4. $F\cos\theta = 50a$ …(1)
$N + F\sin\theta = 10g$ …(2)
$F\cos\theta - 0.6N = 10a$ …(3)
Solving for a, N from (1), (2) and Putting in (3), we get,
$F = \dfrac{30g}{4\cos\theta + 3\sin\theta}$
$\therefore \quad F_{max} = 60 = 15K \ \& \ K = 4.$



5. Segment S_1 has lower wavelength photons, so $E_1 > E_2$ and Intensity is equal, so $n_2 > n_1$.

6. Liquid will experience a force in downward direction always. So, N will be more than Mg. Magnitude of velocity will increase on downward journey and will decrease in upward journey.

7. By energy conservation
$\Rightarrow v = \sqrt{\dfrac{2}{15}}(9.8)\ m/s$
$\omega = 140\sqrt{\dfrac{2}{15}}\ rad/sec$
at bottom point $N - mg = \dfrac{mv^2}{(R-r)}$, where R is radius of curvature at (0, 0)
$\Rightarrow$ N = 49 Newton (approx)

8. voltage sensitivity = $\dfrac{\text{deflection}}{\text{voltage measured}}$

9. $Q = \sigma\left[\pi r^2 + 2\pi rh\right]$
$= 2\times10^{-6}\times\pi\left[\left(\dfrac{2}{10}\right)^2 + 2\left(\dfrac{2}{10}\right)\left(\dfrac{2}{10}\right)\right]$
$= \dfrac{24\pi}{100}\times10^{-6}$
$\simeq 0.72\ \mu C$

10. $U = \dfrac{Q^2}{2C} = \dfrac{\left(72\times10^{-8}\right)^2}{2\times36\times10^{-12}}$
$= 72\times10^{-4}\ J = 7.2\ mJ$

SECTION – C

11. At mean F = kx and at maximum extension spring energy = $\frac{1}{2}ky^2$.

12. Use momentum conservation principle.

13. At resonance $X_L = X_C$ and $Z = R$

So $I_{rms} = \frac{[V_R]_{rms}}{R} = \frac{60}{120} = \frac{1}{2}$ Amp;

$(V_L)_{rms} = I_{rms}X_L \Rightarrow 40 = \frac{1}{2}X_L \Rightarrow X_L = 80\Omega$

$\Rightarrow L = 0.2mH = 0.2\times10^{-3}H$; and $C = 3.125\times10^{-8}F$;

For the current to lag behind the voltage by $\frac{\pi}{4}$;

$$\tan\frac{\pi}{4} = \frac{\omega L - \frac{1}{\omega C}}{R}$$

$\Rightarrow \omega = 8\times10^5 \frac{rad}{sec}$;

14. Use buoyancy principle for tension then calculate frequency.

15. Use formula for Doppler's effect in oblique case.

16. Based on lens formula.

17. $I = \frac{B\ell v}{R_{eq}} = \frac{B\ell a_0 n(R_1+R_2)}{R_1R_2}$ (n is time in second)

So $F_{ext} - F_B = ma_0$

$\Rightarrow F_{ext} = a_0\left[m + \frac{B^2\ell^2(R_1+R_2)n}{R_1R_2}\right]$

$\Rightarrow F_{ext} = \left[1 + \frac{10^{-2}n}{4}\right]$

$\Rightarrow n = 4$;

18. Use the graph to find the value of V_1 for C_3.

19. In Region 1 $K = 2a \Rightarrow \lambda_1 = \frac{h}{\sqrt{4ma}}$

In Region 2 $K = 3a \Rightarrow \lambda_2 = \frac{h}{\sqrt{6ma}}$

$\Rightarrow \frac{\lambda_1}{\lambda_2} = \sqrt{\frac{6}{4}}$

20. According to given condition

$$\left[\frac{9}{5}-1\right]\left[\frac{1}{R}+\frac{1}{R}\right]+\left[\frac{5}{4}-1\right]\left[-\frac{1}{R}-\frac{1}{\infty}\right]=\frac{1}{20}$$

$$\Rightarrow \quad \frac{8}{5R}-\frac{1}{4R}=\frac{1}{20}$$

$$\Rightarrow \quad \frac{27}{20R}=\frac{1}{20} \Rightarrow R=27 \text{ cm}$$

and $\left(\frac{9}{5}-1\right)\left[\frac{1}{R}+\frac{1}{R}\right]+(\mu-1)\left(-\frac{1}{R}-\frac{1}{\infty}\right)=\frac{1}{40}$

$$\Rightarrow \quad \frac{8}{5R}-\frac{\mu-1}{R}=\frac{1}{40}$$

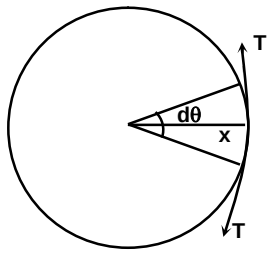
$$\Rightarrow \quad \frac{13-5\mu}{5R}=\frac{1}{40}$$

$$\Rightarrow \quad 13-5\mu=\frac{27}{8}$$

$$\Rightarrow \quad 5\mu=\frac{77}{8} \Rightarrow \mu=\frac{77}{40};$$

SECTION – D

21. Let us suppose that mass per unit length of ring initially is λ; In given condition $0-2T\sin\frac{d\theta}{2}=\lambda R d\theta a$.



$$\Rightarrow \quad a=\frac{-T}{\lambda R} \ \& \ T=\frac{YA}{R}(x-R)$$

$$\Rightarrow \quad a=\frac{-YA}{\lambda R^2}x+\frac{YA}{\lambda R}$$

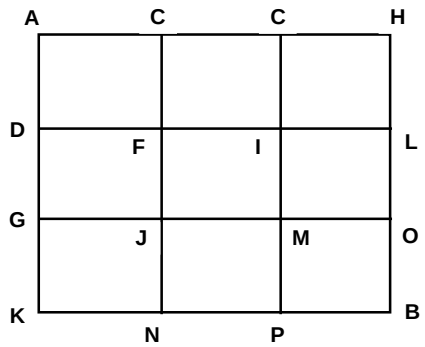
$$\Rightarrow \quad t=\frac{1}{4}\,2\pi\sqrt{\frac{R^2\lambda}{AY}}$$

= 1.25 milisec

22. $A\omega^2 > g$

$$\Rightarrow \quad A>\frac{g}{\omega^2}=\frac{g\lambda^2\mu}{4\pi^2F}=62.5\ \mu m;$$

23. By symmetry of network H; I; J and K are equipotential points.
Besides it potential of D and C; G and E are also same.



Chemistry PART – II

SECTION – A

24. $\frac{P^o - P}{P} = \frac{n}{N} \Rightarrow \frac{W}{M \times N}$

$\Rightarrow \frac{850 - 844.9}{844.9} = \frac{2 \times 76}{M \times 100} \Rightarrow 252$

$n = \frac{252}{32} = 8$

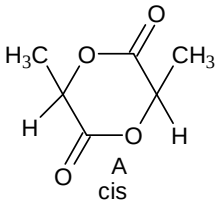
25. Lesser be the gold number, greater be the protective power of the colloid.

27. 4. $\underset{a-x}{A_4(g)} \longrightarrow \underset{4x}{4A(g)}$

At 30 hr, $a - x = 4x \Rightarrow x = \frac{a}{5}$, i.e. 20% A_4 decomposed.

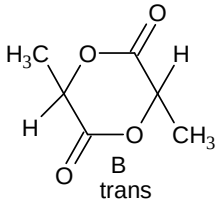
$K \times 30 = \ell n \frac{a}{a-x} = \ell n \frac{5}{4} \Rightarrow t_{1/2} = \frac{30 \ell n 2}{\ell n \frac{5}{4}} \approx 90$ hr

28. $2H_3C-\underset{\text{Lactic acid}}{\underset{OH}{\underset{|}{CH}}-COOH} \longrightarrow$



A
cis

B is optically inactive due to the presence of a point of symmetry.



B
trans

A is optically active, but it is having a C_2 axis, thus, it is dissymmetric.

30. Meq. of oxalic acid $= 40 \times 0.1 \times \frac{250}{100} = 100$

$\therefore$ Eq. weight $= \frac{6.3 \times 1000}{100} = 63$ (hydrated salt.)

Molar mass = 63 × 2 = 126 = 90 + 18 x.

$\Rightarrow$ x = 2.

Equivalent weight of anhydrous acid $= \frac{90}{2} = 45$

Also 20 ml of stock solution $2 \times 40 \times 0.1$

= 8 meq. acid

8 meq. $Ca(OH)_2$ will be present in $= \frac{8}{0.2} = 40$ ml

31. (i) As U is a state function, for any cyclic process, $\Delta U = 0$.

(ii) $\left(\frac{\delta U}{\delta V}\right)_T = \frac{a}{V^2}$.

(iii) During phase transition, T is constant, $dT = 0, \bar{C}_P = \left(\frac{\delta H}{\delta T}\right)_P = \infty$.

(iv) Total molar kinetic energy $= \frac{3}{2}R$.

Internal energy = Total kinetic energy + Potential energy

33. $Cu^{2+} + KI \longrightarrow CuI_2\,(\text{unstable}) \longrightarrow \underset{(\text{white ppt R})}{Cu_2I_2} + I_2 \xrightarrow{\text{excess KI}} CuI_4^{3-}\,(\text{clear solution})(S)$

$$Cu^{2+} + KCN \longrightarrow \underset{\text{unstable}}{Cu(CN)_2} \longrightarrow \underset{\text{white}}{CuCN}\downarrow + \frac{1}{2}(CN)_2\uparrow$$

$$\xrightarrow{\text{excess KCN}} \underset{(\text{clear solution})}{\left[Cu(CN)_4\right]^{3-}} \xrightarrow{Na_2S} \text{No ppt}$$

$$Cu^{2+} + K_3\left[Fe(CN)_6\right] \longrightarrow \underset{(\text{Green})(T)}{Cu_3\left[Fe(CN)_6\right]_2}\downarrow + K^+$$

$$Cu^{2+} + \underset{\text{excess}}{NH_4OH} \longrightarrow \left[\underset{\text{deep blue}}{Cu(NH_3)_4}\right]^{2+}$$

$U = Cu_2\left[Fe(CN)_6\right](\text{chocolate brown})$

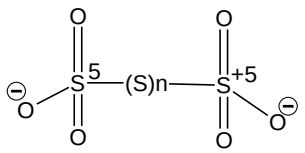
$$SO_4^{2-} + Hg^{2+} + H_2O \longrightarrow \underset{\text{yellow}}{HgSO_4.2H_2O}\downarrow (\text{specific test of } SO_4^{2-})$$

SECTION – C

34. $CH_3 - CH = CH_2 \xrightarrow[(ii)\ H_2O_2/\bar{O}H]{(i)\ B_2H_6/THF} CH_3 - CH_2 - CH_2OH$

$B_2H_6 \longrightarrow 2BH_3$

$$\underset{(3\text{ mole})}{CH_3 - CH = CH_2} \xrightarrow{BH_3} (CH_3CH_2CH_2)_3B \xrightarrow{H_2O_2/\bar{O}H} 3CH_3CH_2CH_2OH$$

35. 

Bond order of S – O bond $= \frac{5}{3}$

Average oxidation state of 'S' atom $\frac{10}{2+n}$.

$\frac{10}{2+n} = \frac{10}{6} \Rightarrow 2 + n = 6 \Rightarrow n = 4.$

36. $\alpha = \frac{10}{200} = 0.05$

$HA \rightleftharpoons H^+ + A^-$

$\left[H^+\right] = C\alpha = 0.2 \times 0.05 = 0.01$

pH = 2

37. $NH_4HS(s) \rightleftharpoons \underset{0.1+P}{NH_3(g)} + \underset{P}{H_2S(g)}$

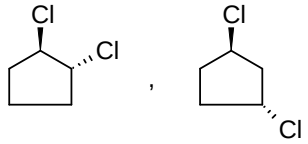
0.1 + 2P = 1.1, thus P = 0.5

$K_P = P_{NH_3}.P_{H_2S} = (0.1+P)\times P$

$= 0.6 \times 0.5 = 0.3$

$= 3 \times 10^{-1}$

Thus, z = 3.

38.  and their enantiomers

39. $\text{Rate} = K[Cl_2][COCl]$

$K_{eq_2} = \dfrac{[COCl]}{[Cl][CO]} \Rightarrow [COCl] = K_{eq_2}[Cl][CO]$

$\dfrac{[Cl]^2}{[Cl_2]} = K_{eq1} \Rightarrow [Cl] = \left\{K_{eq_1}[Cl_2]\right\}^{\frac{1}{2}}$

i.e. $[COCl] = K_{eq_2}K_{eq_1}^{\frac{1}{2}}[CO][Cl_2]^{\frac{1}{2}}$

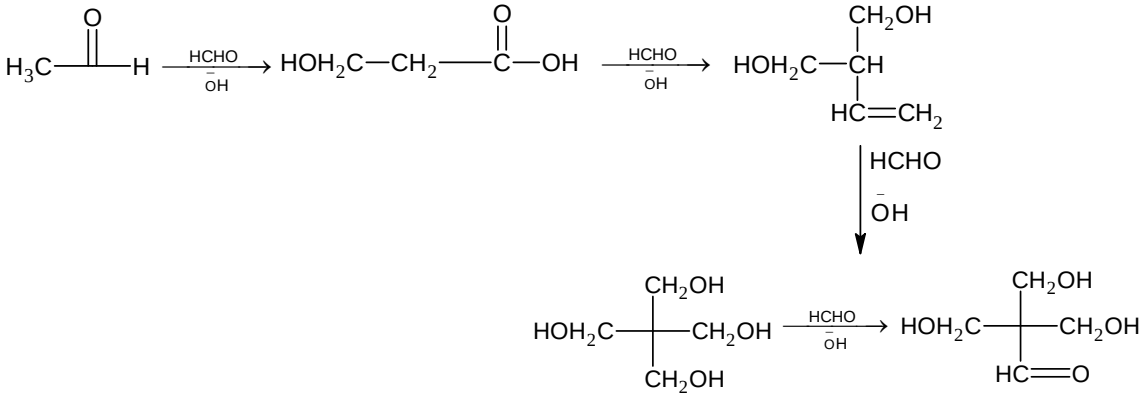
$\text{Rate} = K[Cl_2][COCl]$

$= K[Cl_2]K_{eq_2}K_{eq_1}[CO][Cl_2]^{\frac{1}{2}}$

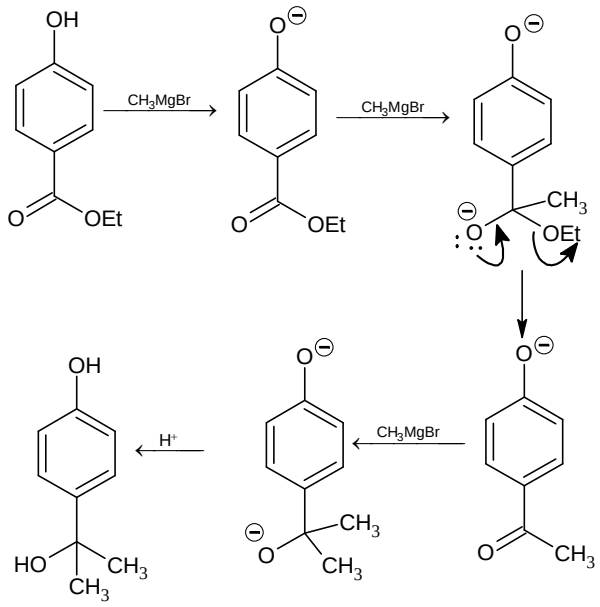
$= K'[CO][Cl_2]^{\frac{3}{2}}$

$\text{Total order} = \dfrac{3}{2} + 1 = \dfrac{5}{2}$

2n = 5

40. 

41.



43. $Ti + 2I_2 \xrightarrow{\Delta} TiI_4$

Mathematics PART – III

SECTION – A

47. $\frac{dy}{dx}-\frac{y}{x}=A$ when $A=\int_1^2 ydx$

I.F = $\frac{1}{x}$

So, $y\cdot\frac{1}{x}=\int A\cdot\frac{1}{x}dx+c=A\ln x+c$

$y = A x \ln x + cx$

Now $f(1) = 2 \Rightarrow 2 = c$

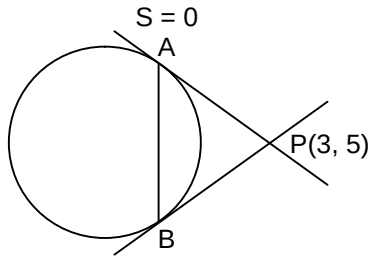
$y = A x \ln x + 2x$

Also, $A=\int_1^2 ydx=\int_1^2 (Ax\ln x+2x)dx$

$= A\left[\ln x\cdot\frac{x^2}{2}-\frac{x^2}{4}\right]_1^2+\left[x^2\right]_1^2 = A\left[\ln 2\cdot 2-1-\left(0-\frac{1}{4}\right)\right]+4-1 = A\left[2\ln 2-\frac{3}{4}\right]+3$

$A\left[\frac{7}{4}-2\ln 2\right]=3;\ A=\frac{12}{7-8\ln 2}$

48. Let circle S = 0 is $x^2 + y^2 + 2gx + 2fy = 0$



Equation of AB as chord of contact

T = 0

$3x + 5y + g(x + 3) + f(y + 5) = 0$

$x(3 + g) + y(5 + f) + 3g + 5f = 0$ ..... (1)

Also, given equation of AB $x + 2y = 3$ ..... (2)

So, $\frac{3+g}{1}=\frac{5+f}{2}=\frac{3g+5f}{-3}$

$6 + 2g = 5 + f$ and $-9 - 3g = 3g + 5f$

$2g - f + 1 = 0$ and $6g + 5f + 9 = 0$

$g=-\frac{7}{8}$ and $f=-\frac{3}{4}$

49. Let foot of perpendicular from (3, –1) on the directrix be (x_1, y_1)

So, $\frac{x_1-3}{2}=\frac{y_1+1}{1}=\frac{-(6-1-3)}{5}$

$\frac{x_1-3}{2}=\frac{y_1+1}{1}=\frac{-2}{5}$

$x_1=3-\frac{4}{5};\ y_1=-1-\frac{2}{5}$

$x_1=\frac{11}{5};\ y_1=-\frac{7}{5}$

Now image of (x_1, y_1) about tangent is focus, which is (x_2, y_2)

$\frac{x_2-\frac{11}{5}}{1}=\frac{y_2+\frac{7}{5}}{1}=\frac{-2\left(\frac{11}{5}-\frac{7}{5}-2\right)}{2}=\frac{6}{5}$

$x_2 = \frac{17}{5}$; $y_2 = \frac{6}{5} - \frac{7}{5} = -\frac{1}{5}$

Focus is $\left(\frac{17}{5}, -\frac{1}{5}\right)$

50. $\vec{a} \times \vec{b} = p\vec{a} + q\vec{b} + r\vec{c}$

Taking dot product with $\vec{a}$, $\vec{b}$, $\vec{c}$

$0 = p + \frac{q}{3} + \frac{r}{3}$ ..... (1)

$0 = \frac{p}{3} + q + \frac{r}{3}$ ..... (2)

$\left[\vec{a}\ \vec{b}\ \vec{c}\right] = \frac{p}{3} + \frac{q}{3} + r$ ..... (3)

$$\left[\vec{a}\ \vec{b}\ \vec{c}\right]^2 = \begin{vmatrix} 1 & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & 1 & \frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} & 1 \end{vmatrix} = 1\left(1 - \frac{1}{9}\right) + \frac{1}{3}\left(\frac{1}{9} - \frac{1}{3}\right) + \frac{1}{3}\left(\frac{1}{9} - \frac{1}{3}\right)$$

$= \frac{8}{9} - \frac{2}{27} - \frac{2}{27} = \frac{8}{9} - \frac{4}{27} = \frac{20}{27}$

From (1) and (2), we get p = q, r = –4q

From (3) $\left[\vec{a}\ \vec{b}\ \vec{c}\right] = \frac{q}{3} + \frac{q}{3} - 4q = -\frac{10q}{3}$

$\frac{20}{27} = \frac{100}{9}q^2 \Rightarrow q^2 = \frac{1}{15}$

51. $I = 2\int_0^{\pi/4} 2^{100} \sec^{100} x\,dx = \int_0^{\pi/4} (2\sec x)^{100}\,dx$

Put sec x + tan x = e^u

sec x(sec x + tan x)dx = e^u du

sec x dx = du

Also, 2 sec x = $e^u + e^{-u}$

$I = 4\int_0^{\ln(\sqrt{2}+1)} \left(e^u + e^{-u}\right)^{99} du$

52. Let $\phi(x) = e^{-2x}f(x)$

$\phi'(x) = e^{-2x}(f'(x) - 2f(x)$

Given $\phi''(x) > 0\ \forall\ x \in \left(0, \frac{1}{2}\right)$ and $\phi'\left(\frac{1}{4}\right) = 0$

$\Rightarrow f'\left(\frac{1}{4}\right) = 2f\left(\frac{1}{4}\right)$; $\phi(0) = \phi\left(\frac{1}{2}\right) = 0$

$\Rightarrow f(x) < 0\ \forall\ x \in \left(0, \frac{1}{2}\right)$

So, $f'\left(\frac{3}{8}\right) - 2f\left(\frac{3}{8}\right) > 0$

$\Rightarrow f'\left(\frac{3}{8}\right) > 2f\left(\frac{3}{8}\right);\ f'\left(\frac{1}{8}\right) < 2f\left(\frac{1}{8}\right)$

53. $\frac{a^2+b^2+c^2}{a+b+c} \geq \left(a^a\ b^b\ c^c\right)^{\frac{1}{a+b+c}}$

$a^2 + b^2 + c^2 \geq a^a\ b^b\ c^c$

Also, $\frac{1}{a}+\frac{1}{b}+\frac{1}{c} \geq 9$

$ab + bc + ac \geq 9abc$

Now $(a + b + c)^2 = 1$

$a^2 + b^2 + c^2 + 2ab + 2ac + 2bc = 1$

$2ab + 2ac + 2bc = 1 - (a^2 + b^2 + c^2)$

$1 - (a^2 + b^2 + c^2) \geq 18\ abc$

$a^2 + b^2 + c^2 \leq 1 - 18\ abc$

54. $n_1 = {}^{2+6-1}C_2 \times 6^4 = {}^{7}C_2 \times 36 \times 36 = 21 \times 36 \times 36 = 27216$

$n_2 = {}^{6+6-1}C_6 = 462$

Now $n_1 = 27216 = 2^4\ 3^5\ 7^1$

n_3 = number of divisors of from $(4k + 1)$ $(k \in N)$ will be 3^2, 3^4, 3×7, $3^3 \times 7$, $3^5 \times 7$ i.e., total 5

55.-56. Let median through A meet BC at D and circle at F

Let GD = x, DF = y then $AG \cdot AF = AB^2$

$2x(3x + y) = 36$ ..... (1)

$xy = 4$

$3x^2 + 4 = 18$

$x^2 = \frac{14}{3}$

So, $AD = 3x = \sqrt{42}$

Also, $AC^2 + AB^2 = 2(AD^2 + BD^2)$

$AC^2 + 36 = 2(42 + 4) = 2 \times 46 = 92$

$AC^2 = 56$

$AC = 2\sqrt{14}$

SECTION – C

57. $f(x) = \frac{\ln\left(\{\sin x\}\{\cos x\}+1\right)}{\{\sin x\}\{\cos x\}}$

$f\left(0^-\right) = \frac{\ln(1\times 1+1)}{1\times 1} = \ln 2$

$f\left(\frac{\pi^+}{2}\right) = \frac{\ln(1\times 1+1)}{1\times 1} = \ln 2$

$f\left(\frac{\pi^-}{2}\right) = \lim_{h\to 0} \frac{\ln\left(\{\cosh\}\{\sinh\}+1\right)}{\{\cosh\}\{\sinh\}} = \lim_{h\to 0} \frac{\ln(\cosh\sinh+1)}{\cosh\sinh} = 1$

58. $I = \int_0^{\pi/4} e^{\sec x} \frac{1}{2}\left[\frac{1+\tan x}{1-\sin x}\right]dx = \frac{1}{2}\int_0^{\pi/4} e^{\sec x}\left(\frac{1+\sin x}{\cos^2 x}+\frac{\tan x(1+\sin x)}{\cos^2 x}\right)dx$

$$= \frac{1}{2}\int_0^{\pi/4} e^{\sec x}\left(\sec^2 x + \tan x \sec x + \tan x \sec^2 x + \tan^2 x \sec x\right)dx$$

$$= \frac{1}{2}\int_0^{\pi/4} e^{\sec x}\left[\sec x \tan x\left(\sec x + \tan x\right) + \left(\sec x \tan x + \sec^2 x\right)\right]dx$$

$$= \frac{1}{2}\left[e^{\sec x}\left(\sec x + \tan x\right)\right]_0^{\pi/4} = \frac{1}{2}\left[e^{\sqrt{2}}\left(\sqrt{2}+1\right) - e\right] = \frac{e^{\sqrt{2}}\left(\sqrt{2}+1\right)}{2} - \frac{e}{2}$$

Here $k = -\frac{e}{2} \Rightarrow 3 \ln (2|k|) = 3$

59. $$\frac{P(T)P\left(\frac{W}{B}\right)}{P(T)P\left(\frac{W}{B}\right) + P(H)P\left(\frac{W}{A}\right)} = \frac{16}{25}$$

$$\frac{\frac{1}{2}\times\left(\frac{4}{6}\right)^n}{\frac{1}{2}\times\left(\frac{4}{6}\right)^n + \frac{1}{2}\left(\frac{3}{6}\right)^n} = \frac{16}{25}$$

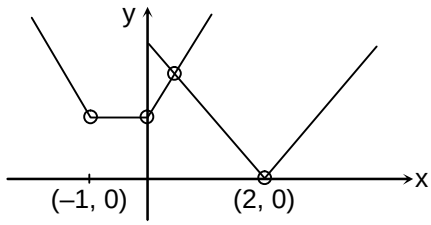
$$\frac{4^n}{4^n + 3^n} = \frac{16}{25} \Rightarrow n = 2$$

60. $f(x) = 2 \min\{|x| + |x+1|, |x-2|\}$

Now, $|x| + |x+1| = 2x + 1, \quad x \geq 0$

$= 1, \quad -1 \leq x < 0$

$= -2x - 1, \quad x < -1$



61. $$|z|^2 - 16 - 3\frac{z^2}{|z|^2} - 3\frac{\bar{z}^2}{|z|^2} + \frac{9}{|z|^2} = 0$$

$$|z|^2 + \frac{9}{|z|^2} - 3\left(\frac{z}{\bar{z}} + \frac{\bar{z}}{z}\right) = 16$$

$$\left(z - \frac{3}{z}\right)\left(\bar{z} - \frac{3}{\bar{z}}\right) = 16$$

$$\left|z - \frac{3}{z}\right| = 4$$

$$\left||z| - \frac{3}{|z|}\right| \leq \left|z - \frac{3}{z}\right| \leq |z| + \frac{3}{|z|}$$

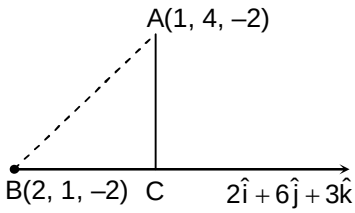
$$\left||z| - \frac{3}{|z|}\right| \leq 4$$

$-4|z| \leq |z|^2 - 3 \leq 4|z|$

$\sqrt{7} - 2 \leq |z| \leq \sqrt{7} + 2$

62. $3^{x-1} = 2x^2 - 4x + 3 = 2(x-1)^2 + 1$ has total 3 solutions
i.e. $x = 1, 2, 3$
Sum of roots = 1 + 2 + 3 = 6

63. $\overrightarrow{BA} = -\hat{i} + 3\hat{j}$



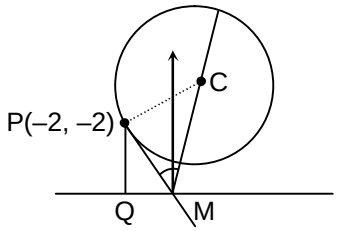
$$AC = \frac{\left|\overrightarrow{BA} \times \left(2\hat{i} + 6\hat{j} + 3\hat{k}\right)\right|}{\left|2\hat{i} + 6\hat{j} - 3\hat{k}\right|} = \frac{\left|-6\hat{k} + 3\hat{j} - 6\hat{k} + 9\hat{i}\right|}{\sqrt{4 + 36 + 9}}$$

$$= \frac{\left|9\hat{i} + 3\hat{j} - 12\hat{k}\right|}{\sqrt{49}} = \frac{\sqrt{81 + 9 + 144}}{7} = \frac{\sqrt{234}}{7} = \frac{3\sqrt{26}}{7}$$

$$p = \frac{3\sqrt{26}}{7}$$

$$\Rightarrow \frac{7p}{\sqrt{26}} = 3$$

64. $\tan 2\alpha = \frac{PC}{PM} = \frac{5}{PM}$ ..... (1)



$$\sin(90 - \alpha) = \frac{PQ}{PM}$$

$$PM = \frac{5/2}{\cos\alpha} = \frac{5}{2\cos\alpha} \quad \text{..... (2)}$$

$$5\cot 2\alpha = \frac{5}{2\cos\alpha}$$

$\Rightarrow 10 \cot 2\alpha \cos\alpha = 5$

65. The equation of tangent at (x_1, y_1)
$\Rightarrow 3xx_1 + 4yy_1 = 1$ ..... (1)
Comparing equation (1) with given equation
$3x + 4y = \sqrt{7}$

$$\frac{x_1}{1} = \frac{y_1}{1} = \frac{1}{\sqrt{7}}$$

$$x_1 = \frac{1}{\sqrt{7}}, y_1 = \frac{1}{\sqrt{7}}$$

66. $16\sin^5 x - 20\sin^3 x + 5\sin x = 16\sin^5 x$

Either $\sin x = 0$ or $\sin^2 x = \frac{1}{4}$

$$\Rightarrow \sin x = \pm\frac{1}{2}$$

$$\Rightarrow x = n\pi \text{ or } x = n\pi \pm \frac{\pi}{6},\ n \in I$$

$$0, \pi, 2\pi, \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$$

Total number of solution = 7

SECTION – D

67. $\tan\theta_1 = \frac{AB}{BD} = \frac{2c}{a}$

$\tan\theta_2 = \frac{BC}{BE} = \frac{2a}{c}$

$\tan\theta_1, \tan\theta_2 = 4$

Angle between median = $\tan^{-1}\frac{3-1}{1+3} = \tan^{-1}\frac{1}{2}$

$\theta = \pi - \tan^{-1}\frac{1}{2}$

Now $\theta_1 + \theta_2 + \theta = \frac{3\pi}{2}$

$\theta_1 + \theta_2 + \pi - \tan^{-1}\frac{1}{2} = \frac{3\pi}{2}$

$\theta_1 + \theta_2 = \frac{\pi}{2} + \tan^{-1}\frac{1}{2}$

$\tan(\theta_1 + \theta_2) = -\cot\tan^{-1}\frac{1}{2} = -2$

$\frac{\tan\theta_1 + \tan\theta_2}{1 - \tan\theta_1\tan\theta_2} = -2$

$\tan\theta_1 + \tan\theta_2 = -2(1-4) = 6$

$\frac{2c}{a} + \frac{2a}{c} = 6 \Rightarrow \frac{c^2 + a^2}{ac} = 3 \Rightarrow \frac{64}{ac} = 3 \Rightarrow ac = \frac{64}{3}$

Area = $\frac{1}{2}ac = \frac{32}{3}$

68. Let α, β, γ are roots then $x^3 - ax^2 + bx + c = (x-\alpha)(x-\beta)(x-\gamma)$

Put $x = 6$

$(6-\alpha)(6-\beta)(6-\gamma) = 3$

Case-I: $6-\alpha = 1, 6-\beta = 1, 6-\gamma = 3$

$\Rightarrow \alpha = 5, \beta = 5, \gamma = 3$

So, $a = 5 + 5 + 3 = 13$

Case-III: $(6-\alpha) = -1, 6-\beta = -1, 6-\gamma = 3$

$\alpha = 7, \beta = 7, \gamma = 3$

$a = 17$

Case-III: $6-\alpha = -1, 6-\beta = 1, 6-\gamma = -3$

$\alpha = 7, \beta = 5, \gamma = 9$

$a = 21$

69. The number of ways = D(4) + 2D(3) + D(2) = 9 + 4 + 1 = 14 = k then $\frac{k}{3} = 4.66$